

QUIZ 3 -MTH301-page1-4

● Graded

Student
YATIN PREETAM BHOJWANI

Total Points
10 / 12 pts

Question 1

Q1

4 / 6 pts

+ 0 pts Completely Incorrect

+ 6 pts Completely correct

✓ + 2 pts Part (a) is correct.

✓ + 2 pts Finding correct fixed point (0,0) in part (b).

+ 2 pts Proof of strict contraction is correct in part (b).

+ 4 pts Part (b) is correct.

1 How?

Question 2

Q2

Resolved 6 / 6 pts

2 (a)

+ 0 pts 2 (a) Completely Incorrect

✓ + 4 pts 2(a) Completely Correct

+ 1 pt 2 (a) Partially Correct

+ 2 pts 2 (a) Partially Correct

+ 3 pts 2 (a) Partially Correct

2 (b)

+ 0 pts 2 (b) Completely Incorrect

✓ + 2 pts 2 (b) Completely Correct

+ 1 pt 2 (b) Partially Correct

🔄 Regrade Request

Submitted on: Nov 05

In part a) one mark was cut because of $x-1/n^2 > x-1/N^2$ because $n>N$ was specified, but it was actually outside the box that gradescope automatically makes , so it wasn't visible to the grader. I am attaching the Google drive link showing that it is indeed written, i hope the request is considered

https://drive.google.com/file/d/1G29byvNgX7u_3dyaD9r9xHjiB-kcKfile/view?usp=drivesdk

Pls open and see the fact that I have written $n>N$ just below the underlined statement

Done.

Reviewed on: Nov 09

ANALYSIS - I : MTH 301 : QUIZ : 2025-26

Name:

YANN PREETAM BHOJWANI

Roll No:

241211

• Write your name cleanly in **CAPITAL** letters and roll number inside the boxes.

• Write answers in the space provided only. Maximum marks: 25 Time: 18:30 - 19:45.

1. (a) Find an example of a space M and a *strict contraction* map $F : M \rightarrow M$ having no fixed point where M is *not* complete.
- (b) Let $F : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a map defined for any $(x_1, x_2) \in \mathbb{R}^2$ by $F(x_1, x_2) = \frac{1}{4}(x_1 + x_2, x_1 + 2x_2)$. Show that F is a *strict contraction* and find the unique fixed point. [2+4]

(a) ~~$M = [0, 1]$~~ $M = (0, 1)$
is an incomplete space
and $f(x) = \frac{x}{2}$.

also $|f(y) - f(x)| = \left| \frac{y}{2} - \frac{x}{2} \right|$
 $\leq \frac{1}{2} |y - x|$

(a) take $M = (0, 1)$

this is incomplete space
{ $x_n : \frac{1}{n}$ is Cauchy but does not converge}

let $F : M \rightarrow M$
 $F(x) = \frac{x}{2}$

show F is strict contraction

$$|F(y) - F(x)| = \left| \frac{y}{2} - \frac{x}{2} \right| = \frac{1}{2} |y - x| \leq \frac{1}{2} |y - x|$$

$$\Rightarrow |F(y) - F(x)| \leq \frac{1}{2} |y - x|$$

(b) $F : \mathbb{R}^2 \rightarrow \mathbb{R}^2$

$$F(x_1, x_2) = \frac{1}{4}(x_1 + x_2, x_1 + 2x_2)$$

take $y, x \in \mathbb{R}^2$

$y = (y_1, y_2)$, $x = (x_1, x_2)$
to show it is

$$\|F(y) - F(x)\| = \left\| \frac{1}{4}(y_1 + x_1, y_2 + 2x_2) - \frac{1}{4}(x_1 + x_2, x_1 + 2x_2) \right\|$$

to show it is contraction

$$d(y, x) \quad d(F(y), F(x)) \leq \alpha d(y, x)$$

here d is the $\|\cdot\|$ norm

$$\left\| \frac{1}{4}(y_1 + x_1, y_2 + 2x_2) - \frac{1}{4}(x_1 + x_2, x_1 + 2x_2) \right\|$$

$$= \frac{1}{4} \sqrt{(y_1 + x_1 - x_1 - x_2)^2 + (y_2 + 2x_2 - x_1 - 2x_2)^2}$$

$$= \frac{1}{4} \sqrt{(y_1 - x_2)^2 + (y_2 - x_1)^2}$$

$$= \frac{1}{4} \sqrt{(y_1 - x_1)^2 + (y_2 - x_2)^2 + 2(y_1 - x_1)(y_2 - x_2) + (y_1 - x_1)^2 + (2y_2 - 2x_2)^2 + 2(y_1 - x_1)(2y_2 - 2x_2)}$$

$$= \frac{1}{4} \sqrt{2(y_1 - x_1)^2 + 5(y_2 - x_2)^2 + 6(y_1 - x_1)(y_2 - x_2)}$$

$$= \frac{1}{4} \sqrt{(y_1 - x_1)^2 + \frac{5}{2}(y_2 - x_2)^2 + 3(y_1 - x_1)(y_2 - x_2)}$$

$$= \frac{1}{4} \sqrt{\frac{1}{8}(y_1 - x_1)^2 + \frac{5}{16}(y_2 - x_2)^2 + \frac{6}{16}(y_1 - x_1)(y_2 - x_2)}$$

$$= \sqrt{\frac{1}{8}(y_1 - x_1)^2 + \frac{5}{16}(y_2 - x_2)^2 + \frac{6}{16}(y_1 - x_1)(y_2 - x_2)}$$

$$\sqrt{(y_1 - x_1)^2 + (y_2 - x_2)^2} = \|y - x\|$$

(as $\frac{1}{8}(y_1 - x_1)^2 + \frac{5}{16}(y_2 - x_2)^2 + \frac{6}{16}(y_1 - x_1)(y_2 - x_2) < (y_1 - x_1)^2 + (y_2 - x_2)^2$)

$\Rightarrow \|f(y) - f(x)\| < \|y - x\|$
hence a strict contraction.

finding fixed pt. (unique)

$$F(x_1, x_2) = (x_1, x_2)$$

$$\frac{1}{4}(x_1 + x_2, x_1 + 2x_2) = (x_1, x_2)$$

$$\begin{cases} x_1 = x_1 + x_2 \\ x_2 = x_1 + 2x_2 \end{cases} \Rightarrow \begin{cases} x_1 = 0 \\ x_2 = 0 \end{cases}$$

fixed pt is $(0, 0)$

..... Rough Work

fixed pt. strict contraction mapping & complete space.
if $\|f(x) - f(y)\| < \alpha d(x, y)$

if i take x_0
($f^n(x_0)$) is Cauchy
 $x \Rightarrow (f^n(x_0))$
 $\lim_{n \rightarrow \infty} f^n(x_0)$

$$\frac{1}{4}(x_1 + x_2, x_1 + 2x_2)$$

$$2[y_1^2 + x_1^2 - 2y_1x_1] + 5[y_2^2 + x_2^2 - 2y_2x_2] + 6y_1y_2 - 6y_1x_2 - 6x_1y_2 + 6x_1x_2$$

$$2y_1^2 + 2x_1^2 - 4y_1x_1 + 5y_2^2 + 5x_2^2 - 10y_2x_2 + 6y_1y_2 - 6y_1x_2$$

eqn: $x = \frac{n}{2}$ has no roots in $[0,1]$

1

$$x^2 - 2x_2^2$$

$$\frac{1}{4} (x_1 + 2x_2)^2 = x_2^2$$

$$\frac{1}{4} (x_1 + x_2)^2 = x_2^2$$

$$x_1 + x_2 = 4x_2$$

$$x_1 + x_2 = 4x_1$$

$$= (x_1, x_2)$$

$$2y_1^2 + 5y_2^2 + 2x_1^2 + 5x_2^2 - 10y_1$$

$$+ 6x_1x_2$$

2. (a) Let $g: \mathbb{R} \rightarrow \mathbb{R}$ be uniformly continuous, and define $f_n(x) := g\left(x - \frac{1}{n^2}\right)$. Show that f_n uniformly converges to g on $\mathbb{R}$.

(b) Let $f: [0, 3] \rightarrow \mathbb{R}$ be a continuous function. Justify whether $\lim_{n \rightarrow \infty} \int_{\frac{1}{n}}^{2+\frac{1}{n}} f(y) dy = \int_0^2 f(y) dy$.

[4+2]

(a) $g: \mathbb{R} \rightarrow \mathbb{R}$ is uniformly continuous.

$$\forall \epsilon > 0, \exists \delta_\epsilon > 0 \text{ s.t. } |y - x| < \delta_\epsilon \\ \Rightarrow |f(y) - f(x)| < \epsilon$$

$$\text{and that } n - \frac{1}{n} < n + \frac{1}{n} \\ \forall n > N_\epsilon$$

(b) $\lim_{n \rightarrow \infty} \int_{\frac{1}{n}}^2 f(y) dy$ as f is continuous integral exists.

By Fundamental Theorem of Calculus. (Riemman integral)

$$\int_a^b f(y) dy = F(b) - F(a)$$

... continuous

Yes it is true.